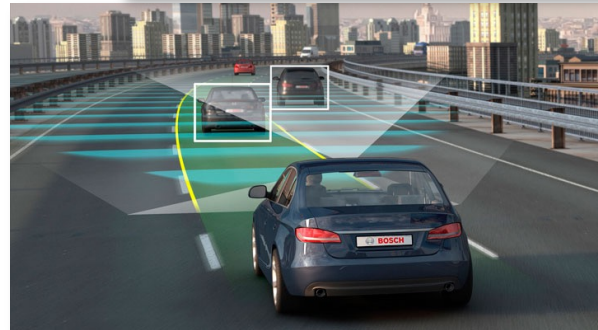
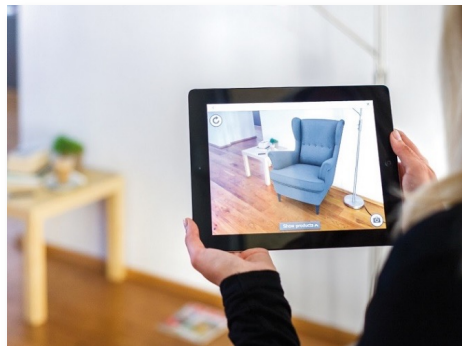
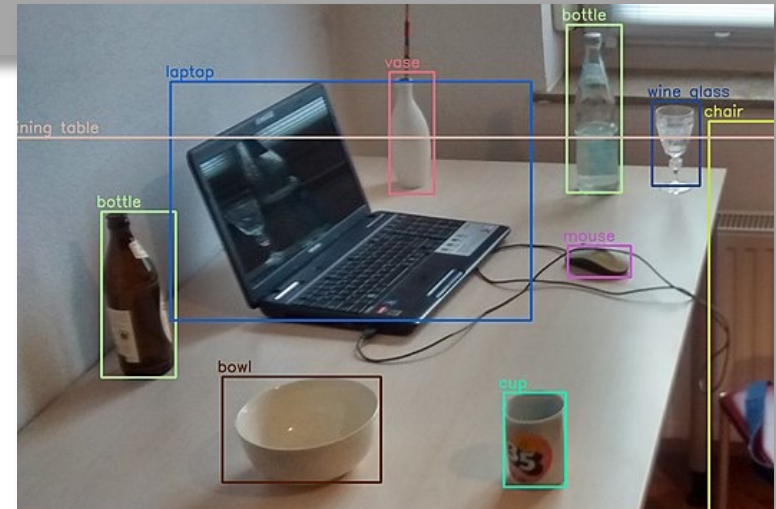
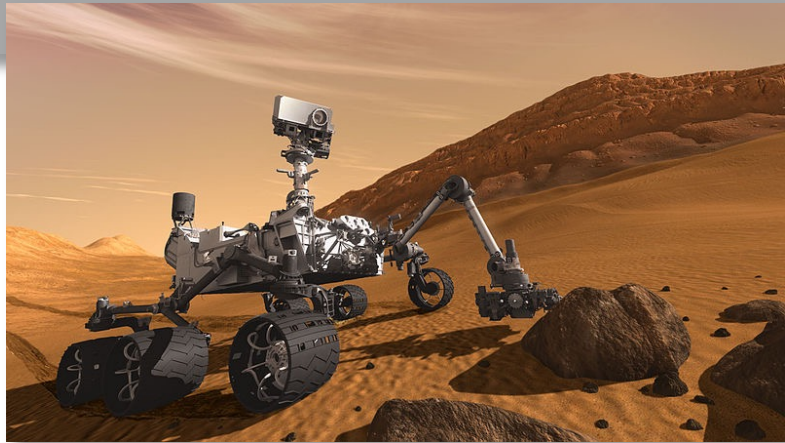


CS5670: Intro to Computer Vision (Cornell Tech)

Instructor: Noah Snavely

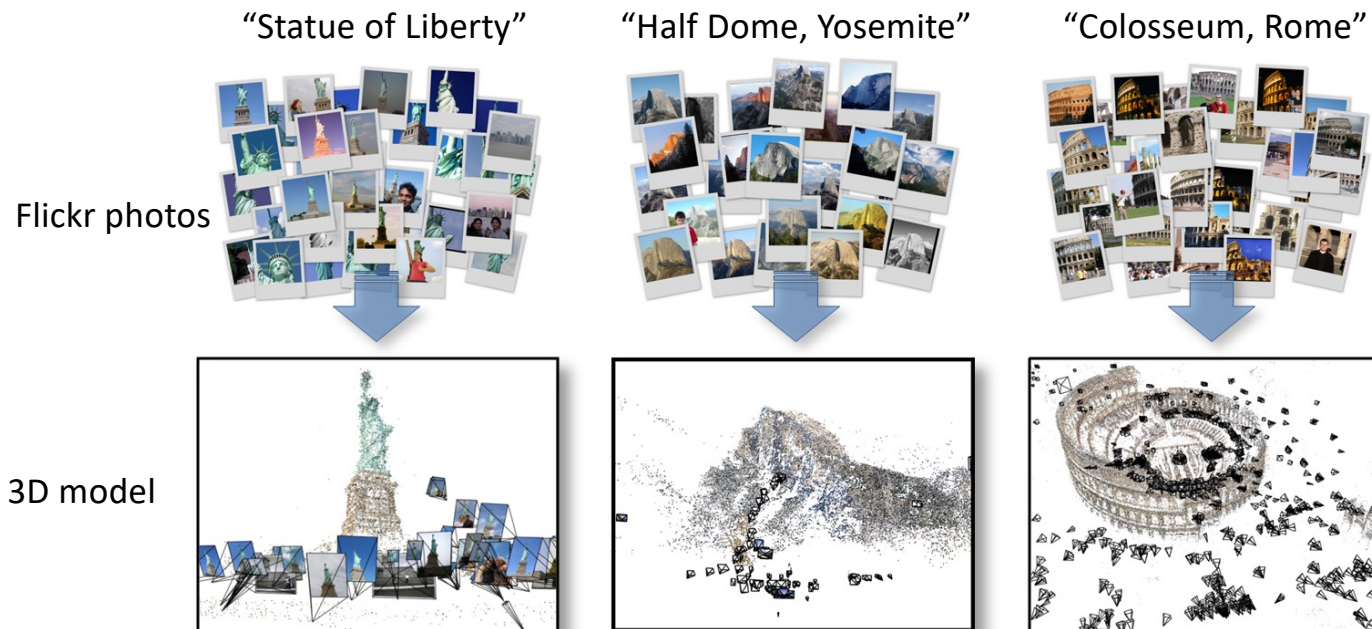


Instructor

- Noah Snavely (snavely@cs.cornell.edu)
- Research interests:
 - Computer vision and graphics
 - 3D reconstruction and visualization of Internet photo collections
 - Deep learning for computer graphics

Noah's work

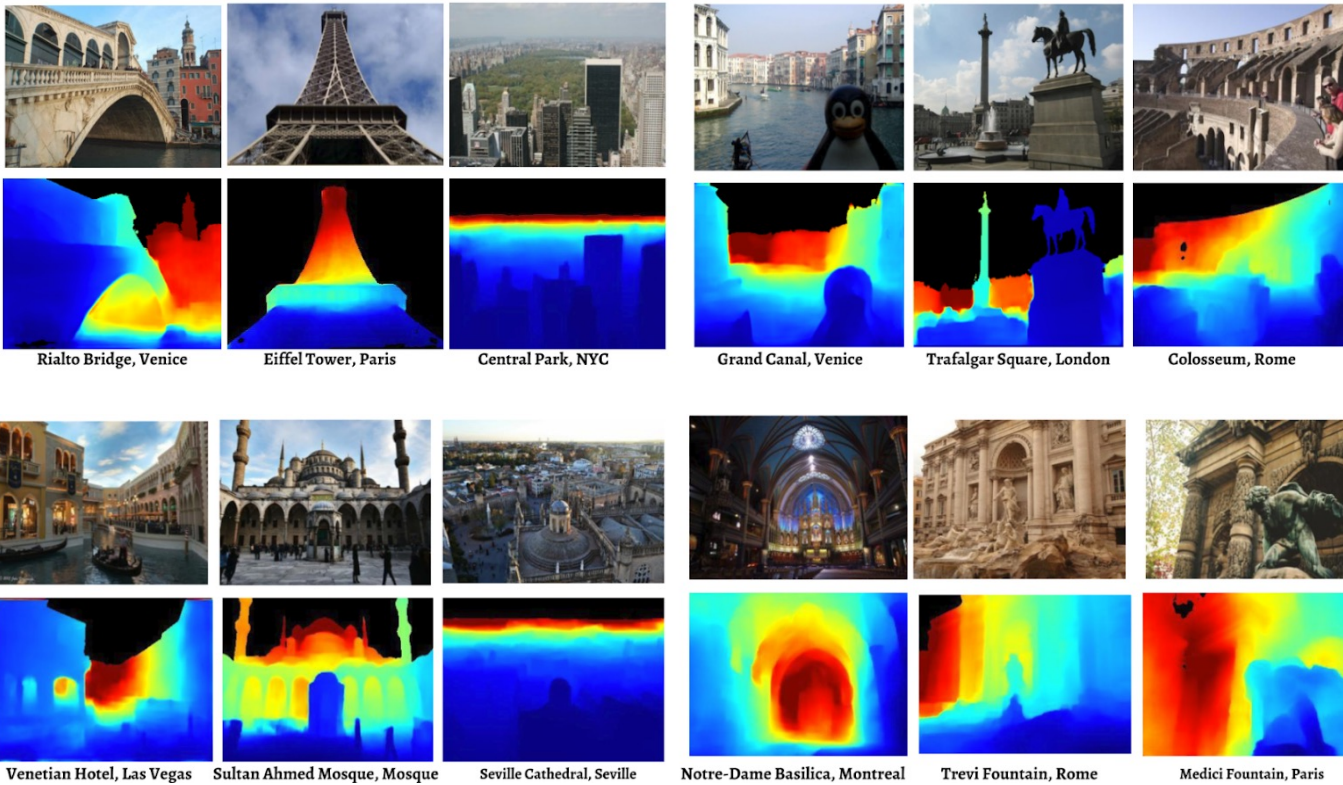
- Automatic 3D reconstruction from Internet photo collections



City-scale 3D reconstruction

Reconstruction of Dubrovnik, Croatia, from ~40,000 images

Depth from a single image



Visualizing scenes from tourist photos



Reconstructing dynamic 3D scenes



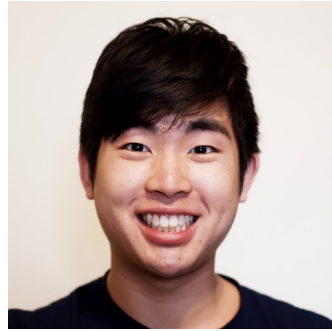
DynIBaR: Neural Dynamic Image-Based Rendering [<https://dynibar.github.io/>]

Zhengqi Li, Qianqian Wang, Forrester Cole, Richard Tucker, Noah Snavely
CVPR 2023

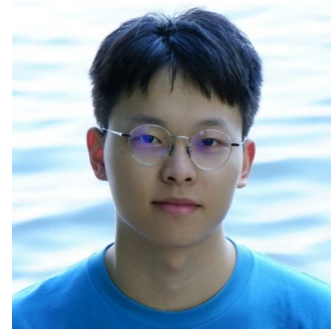
Teaching assistants



Ruojin Cai
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Haian Jin
hj453@cornell.edu

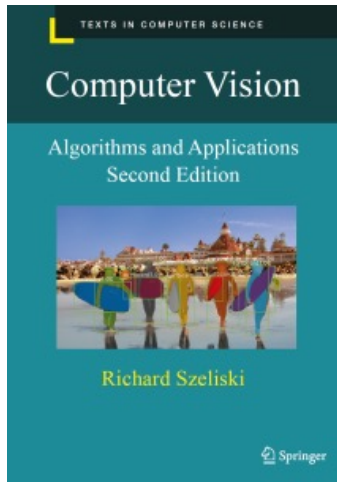


Amritansh Kwatra
ak2244@cornell.edu

- Please check back on course webpage for office hours

<https://www.cs.cornell.edu/courses/cs5670/2025sp/>

Important information



- Textbook:
Rick Szeliski, *Computer Vision: Algorithms and Applications* online at:
<http://szeliski.org/Book/>
- Course webpage:
<http://www.cs.cornell.edu/courses/cs5670/2025sp/>
- Canvas Page:
<https://canvas.cornell.edu/courses/61359>
- Announcements/discussion via Ed Discussions (via Canvas)
- Assignment turnin via GitHub Classroom and CMSX:
<https://cmsx.cs.cornell.edu>

Today

1. What is computer vision?
2. Why study computer vision?
3. Course overview
4. Images & image filtering [time permitting]

Today

- Readings
 - Szeliski, Chapter 1 (Introduction)

Every image tells a story



- Goal of computer vision: perceive the “story” behind the picture
- Compute properties of the world
 - 3D shape
 - Names of people or objects
 - What happened?

The goal of computer vision



0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Can computers match human perception?



- Yes and no
 - humans are better at “hard” things, are more robust, and make inferences quickly and cheaply
- But huge progress
 - Accelerating in the last 10 years due to deep learning, large vision-language models
 - What is considered “hard” keeps changing



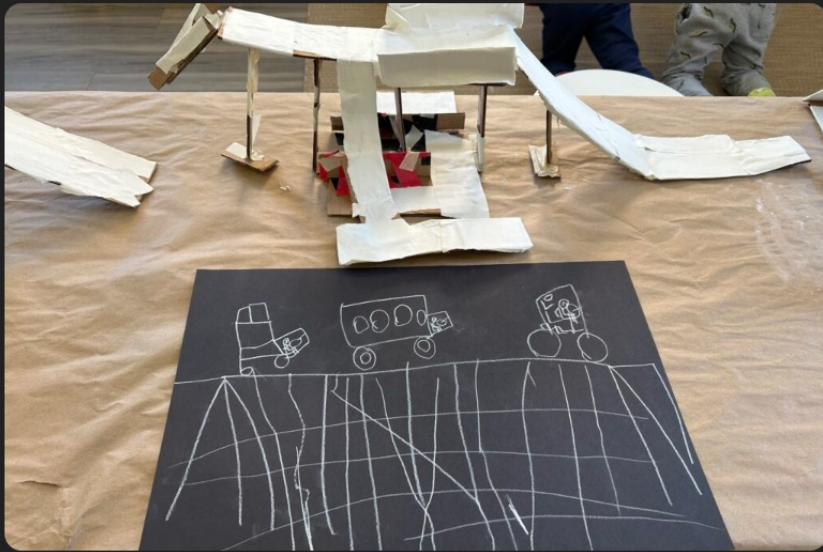
La Gare Montparnasse, 1895

What do you see in this picture? What do you think happened before this picture was taken?



In the picture, we see the famous Montparnasse derailment from 1895. It shows a steam locomotive that has crashed through the station wall and is hanging precariously over the street below. The photograph captures the aftermath of a dramatic accident where the train failed to stop while approaching the buffer stop at the Gare Montparnasse terminal in Paris.

Before this picture was taken, the train, arriving from Granville to Paris, was running late. The driver, trying to make up for lost time, was likely going too fast and unable to stop the train in time upon reaching the station. The train overran the buffer stop, crossed the station concourse, crashed through a 60 cm thick wall, and ended up nosediving onto the Place de Rennes below. Remarkably, only one person, a woman selling newspapers on the street, was killed by falling masonry, and several others were injured. This image has become a symbol of dramatic accidents and is frequently used to illustrate the need for careful management of technology and speed.



What do you see in this picture?



In this image, you can see a creative and playful representation of a bridge construction made from paper materials. There are several sections of the bridge, including ramps and supports, all crafted from white and red paper, arranged over a brown paper surface. Beside this bridge model, there is a black paper with white chalk drawings that depict vehicles like trucks and motorcycles, perhaps to simulate a traffic scenario on the bridge. It appears to be a hands-on educational activity, possibly designed to teach children about engineering, construction, and physics through interactive play. The scene is set in a room where children are present, indicating that this might be a classroom or workshop setting.

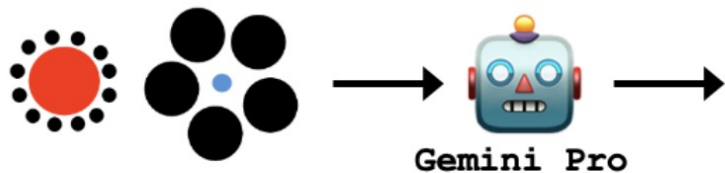


Current models still make very silly mistakes

Prompt: "Which is bigger, the blue circle or the red circle?"



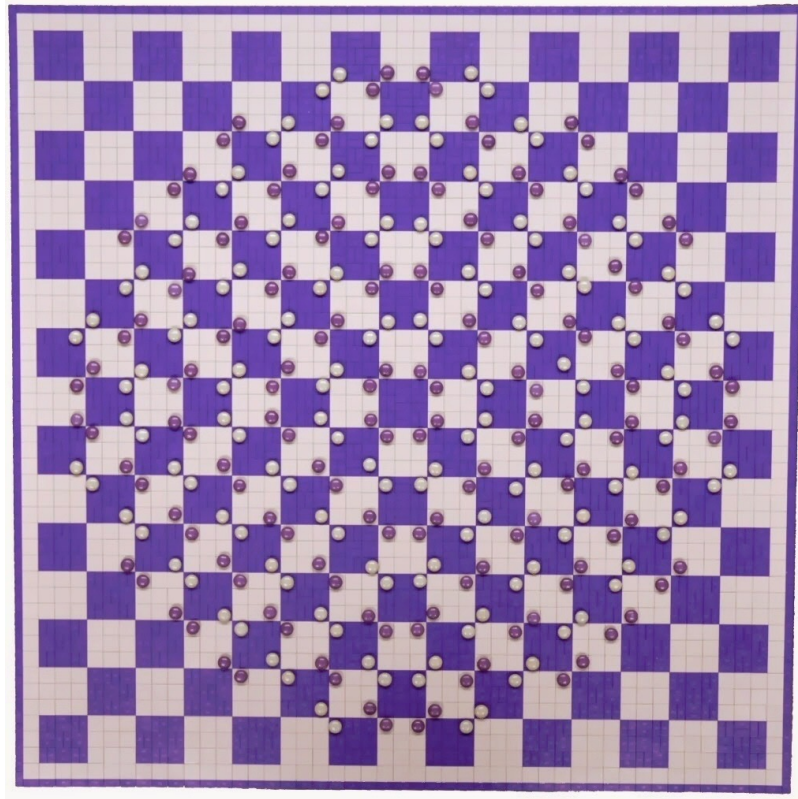
"This is a classic optical illusion! The red and blue circles are actually the same size. Our brains tend to perceive the red circle as smaller because it's surrounded by larger black circles. The blue circle, surrounded by smaller black circles, appears larger in comparison."



"This is a classic optical illusion! The red and blue circles are actually the same size. Our brains tend to perceive the red circle as larger because it's surrounded by smaller circles, while the blue circle is surrounded by larger circles. This is an example of the Ebbinghaus illusion."

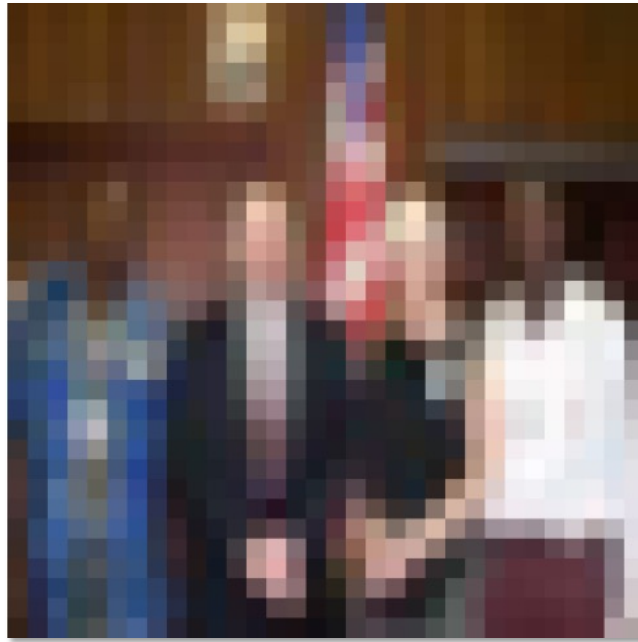
[Tomer Ullmann, *The Illusion-Illusion: Vision Language Models See Illusions Where There are None*, arXiv 2024]

Human perception has its shortcomings



<https://twitter.com/pickover/status/1460275132958662657/>

But humans can tell a lot about a scene from a little information...



Source: "80 million tiny images" by Torralba, et al.



The goal of computer vision

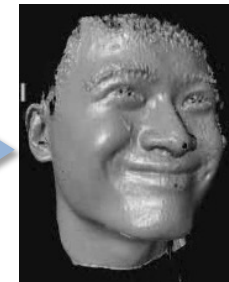
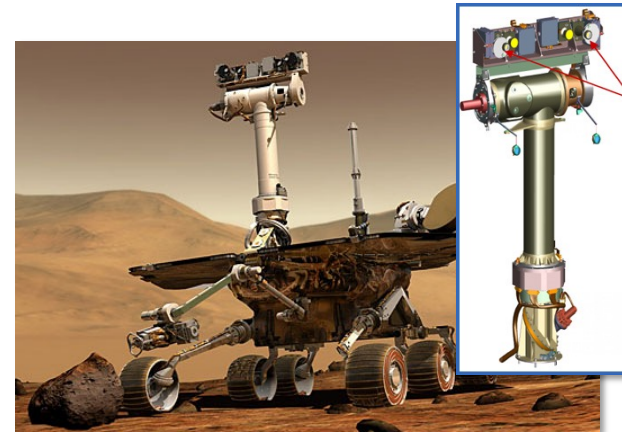
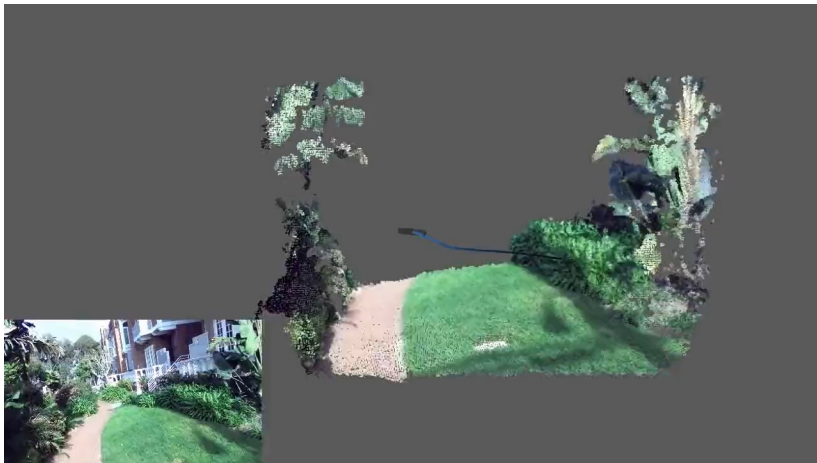


The goal of computer vision

- Compute the 3D shape of the world

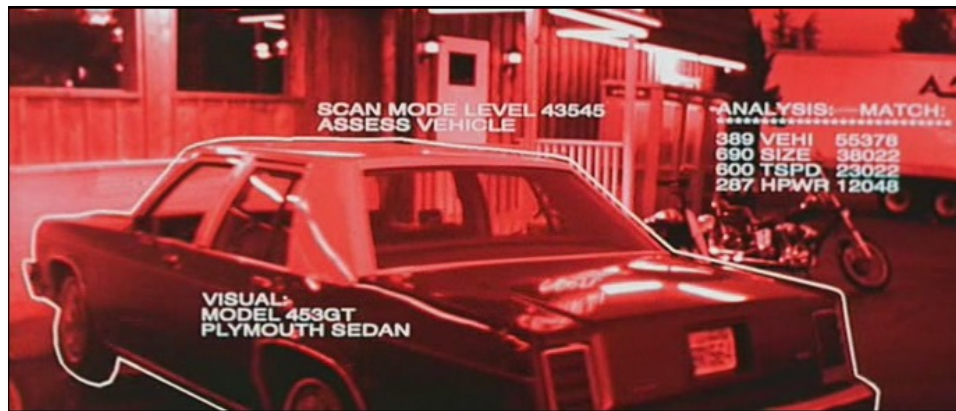


ZED 2i Camera



The goal of computer vision

- Recognize objects and people



Terminator 2, 1991





slide credit: Fei-Fei, Fergus & Torralba

The goal of computer vision

- “Enhance” images





The goal of computer vision

- Forensics



Source: Nayar and Nishino, "Eyes for Relighting"



Source: Nayar and Nishino, "Eyes for Relighting"



Source: Nayar and Nishino, "Eyes for Relighting"

The goal of computer vision

- Improve photos (“Computational Photography”)



Super-resolution (source: 2d3)



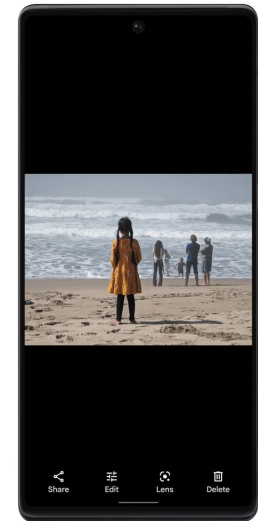
Low-light photography

(credit: [Hasinoff et al., SIGGRAPH ASIA 2016](#))



Depth of field on cell phone camera

(source: [Google Research Blog](#))



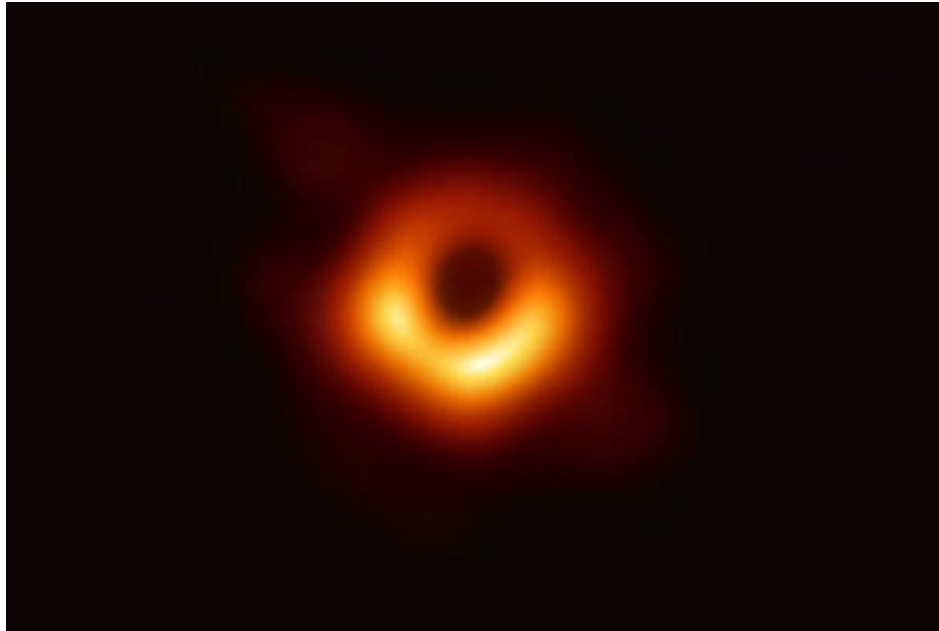
Removing objects
([Google Magic Eraser](#))

The New York Times

***Darkness Visible, Finally:
Astronomers Capture First Ever Image
of a Black Hole***

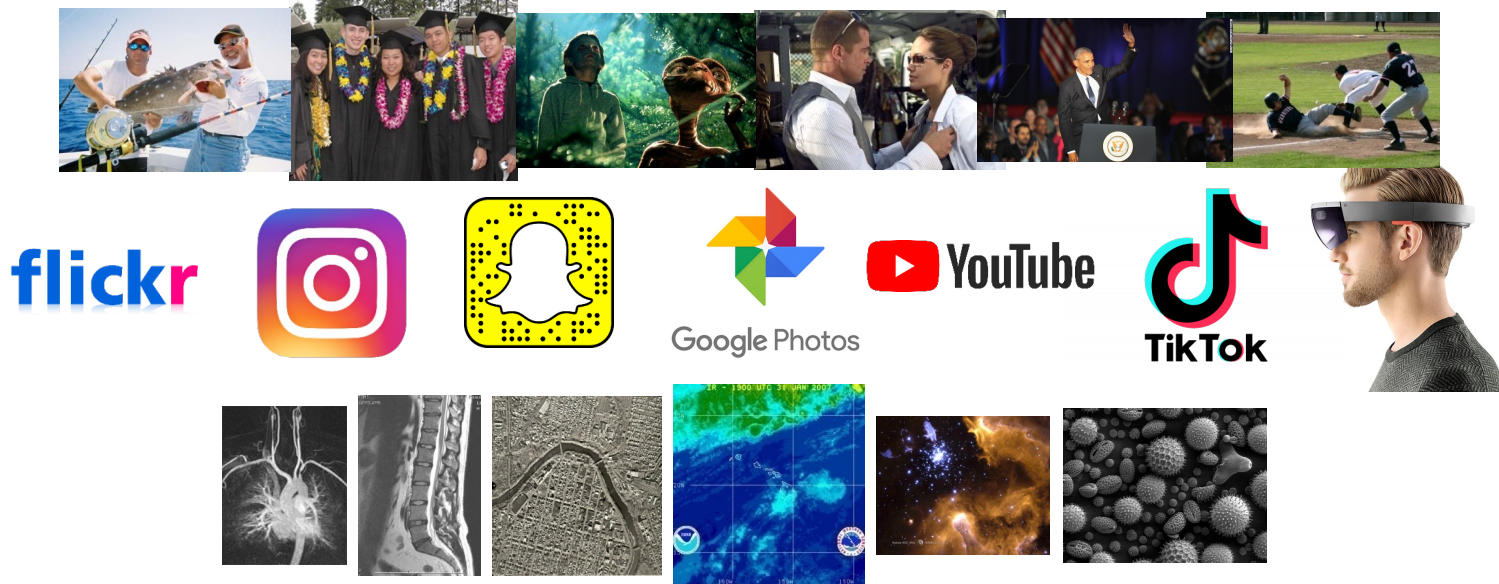
Astronomers at last have captured a picture of one of the most secretive entities in the cosmos.

April 10, 2019



Why study computer vision?

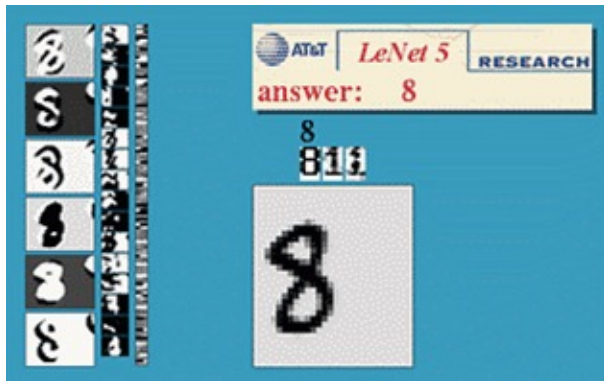
- Billions of images/videos captured per day



- Huge number of potential applications
- The next slides show the current state of the art

Optical character recognition (OCR)

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs (1990's)

<http://yann.lecun.com/exdb/lenet/>



License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



Automatic check processing



Sudoku grabber

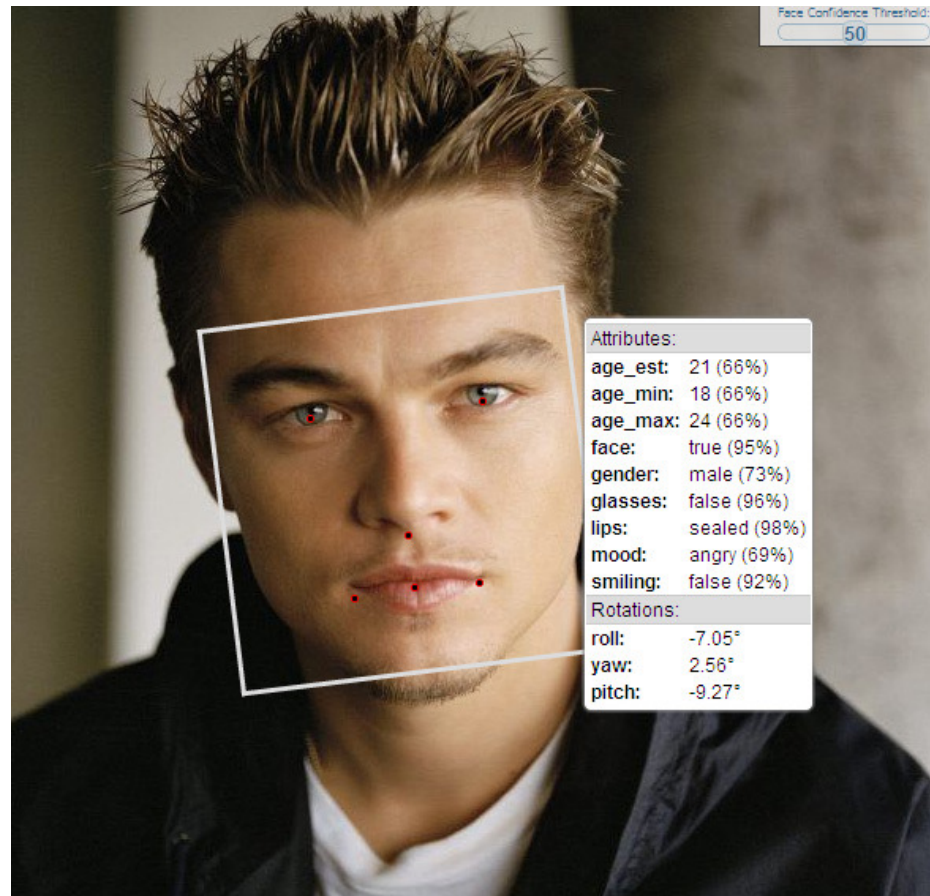
<http://sudokugrab.blogspot.com/>

Face detection

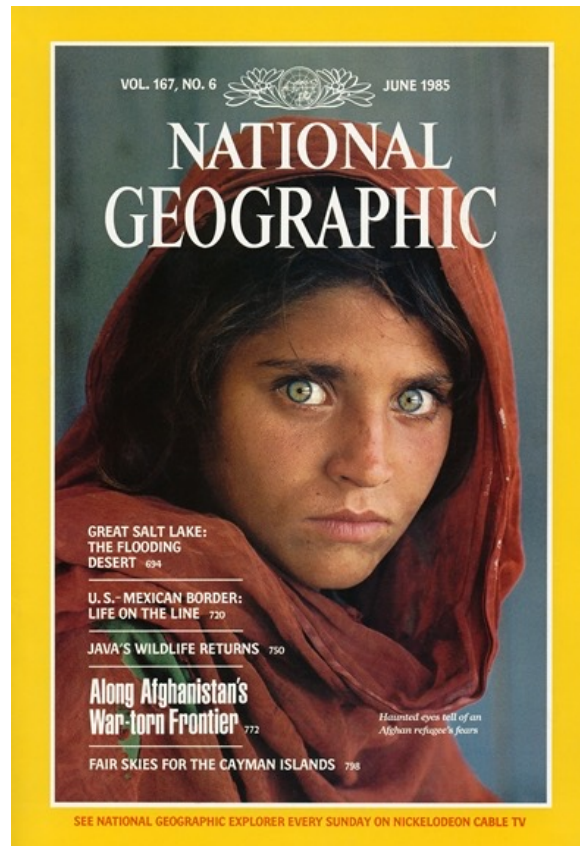


- Nearly all cameras detect faces in real time
 - (Why?)

Face analysis and recognition



Vision-based biometrics



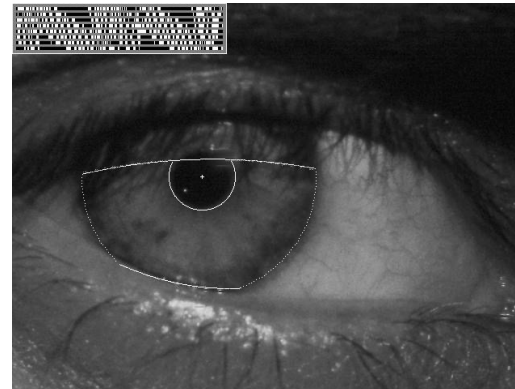
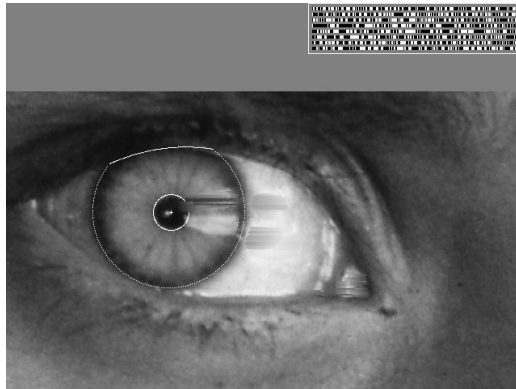
Who is she?

Source: S. Seitz

Vision-based biometrics

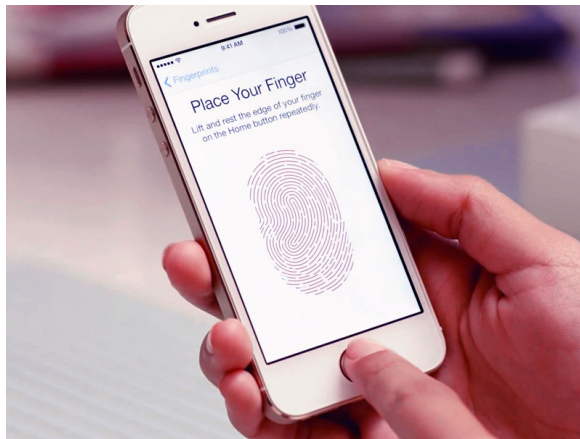


"How the Afghan Girl was Identified by Her Iris Patterns" Read the [story](#)



Source: S. Seitz

Login without a password



Fingerprint scanners on many new smartphones and other devices



Face unlock on Apple iPhone X
See also <http://www.sensiblevision.com/>



The Secretive Company That Might End Privacy as We Know It

A little-known start-up helps law enforcement match photos of unknown people to their online images — and “might lead to a dystopian future or something,” a backer says.



New York Times, Jan. 18, 2020
by Kashmir Hill

Researchers warn peace sign photos could expose fingerprints

But the likelihood of anyone actually using images to recreate prints is pretty slim.



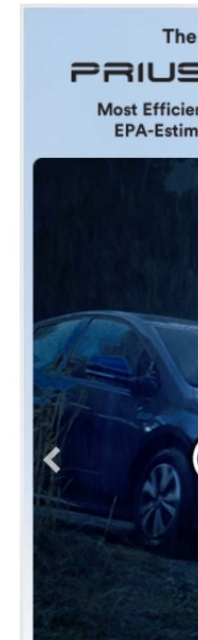
Jamie Rigg, @jmerigg
01.13.17 in [Security](#)

Comments

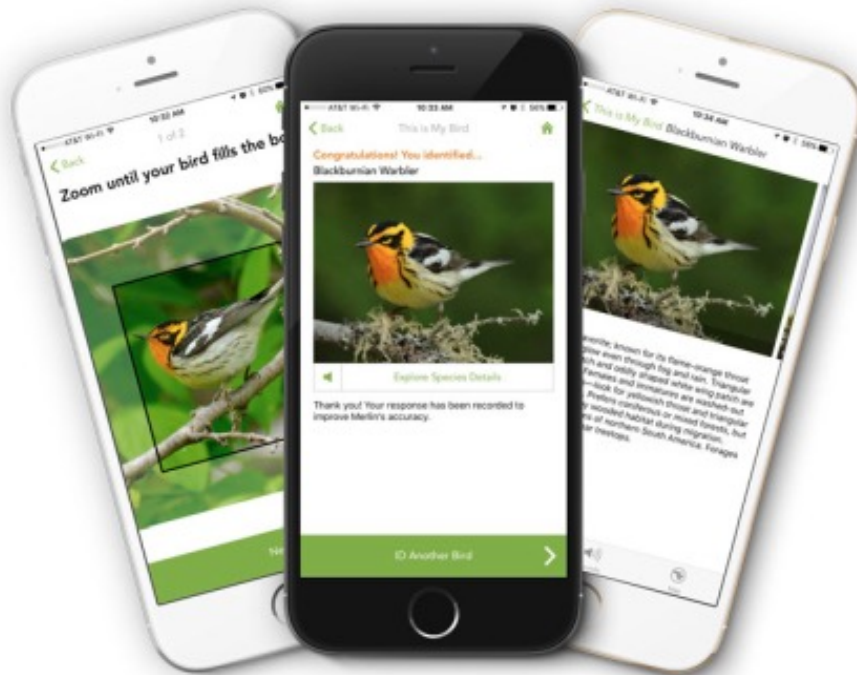
1721
Shares



Getty



Bird identification



Merlin Bird ID (based on Cornell Tech technology!)

Special effects: shape capture



The Matrix movies, ESC Entertainment, XYZRGB, NRC

Source: S. Seitz

Special effects: motion capture



Pirates of the Caribbean, Industrial Light and Magic

Source: S. Seitz

MOVIES



Robert De Niro said no green screen. No face dots. How ‘The Irishman’s’ de-aging changes Hollywood



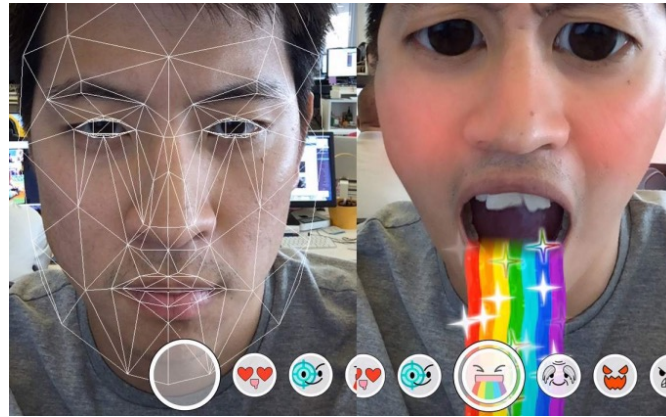
Makeup and wig work got Robert De Niro partway to his character, Frank Sheeran, at 41, left. It took a specially built camera and visual artists to get all the way there, as before-and-after images show. (Netflix)

By JOSH ROTTENBERG | STAFF WRITER JAN. 2, 2020

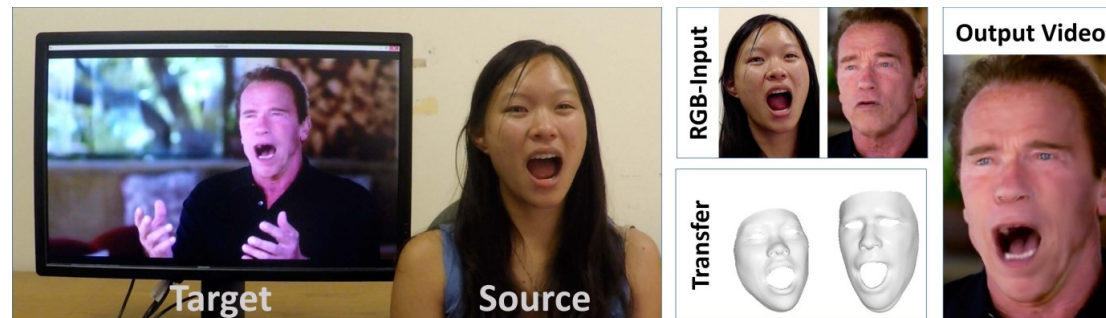
Los Angeles Times



3D face tracking w/ consumer cameras



Snapchat Lenses



[Face2Face system](#) (Thies et al.)

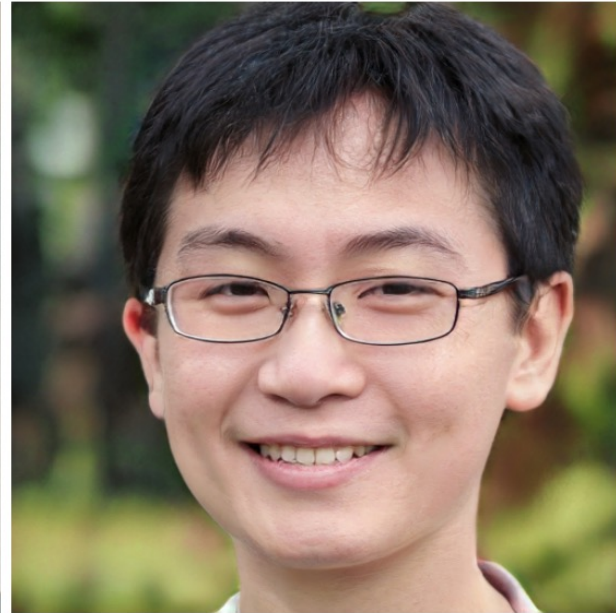
Image synthesis



Karras, et al., *Progressive Growing of GANs for Improved Quality, Stability, and Variation*, ICLR 2018

Which face is real?

Click on the person who is real.



<https://www.whichfaceisreal.com/>

Image synthesis



"An astronaut riding a horse in a photorealistic style" – DALL-E 2



"A photo of a Corgi dog riding a bike in Times Square. It is wearing sunglasses and a beach hat" – Imagen

Sports



Sportvision first down line

[Explanation](http://www.howstuffworks.com) on www.howstuffworks.com



Source: S. Seitz

Smart cars

The screenshot displays the Mobileye website with a central banner titled "Our Vision. Your Safety." featuring a car with four camera fields of view: rear, forward, and two side cameras. Below the banner are three product highlights: "EyeQ Vision on a Chip" with an image of the chip, "Vision Applications" showing a pedestrian detection box, and "AWS Advance Warning System" with a circular display showing a car icon and a distance of 0.8. To the right, a "News" sidebar lists articles about Volvo's collision warning system, and an "Events" sidebar lists Mobileye's presence at Equip Auto in Paris and SEMA in Las Vegas.

manufacturer products consumer products

Our Vision. Your Safety.

rear looking camera forward looking camera side looking camera

➤ **EyeQ** Vision on a Chip

➤ **Vision Applications**
Road, Vehicle, Pedestrian Protection and more

➤ **AWS** Advance Warning System

News

- Mobileye Advanced Technologies Power Volvo Cars World First Collision Warning With Auto Brake System
- Volvo: New Collision Warning with Auto Brake Helps Prevent Rear-end

Events

- Mobileye at Equip Auto, Paris, France
- Mobileye at SEMA, Las Vegas, NV

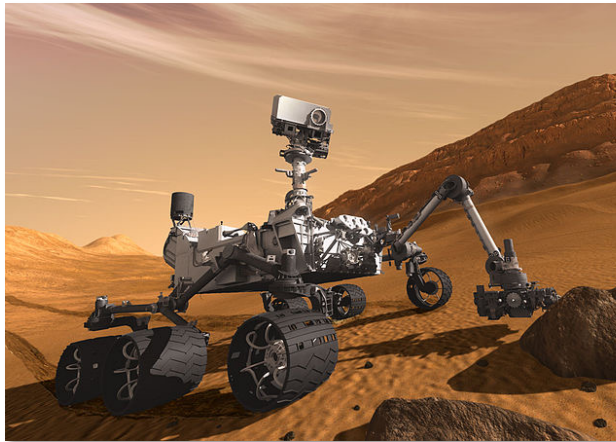
- [Mobileye](#)
- Tesla Autopilot
- Safety features in many cars

Self-driving cars



Waymo

Robotics



NASA's Mars Curiosity Rover

[https://en.wikipedia.org/wiki/Curiosity_\(rover\)](https://en.wikipedia.org/wiki/Curiosity_(rover))



Amazon Picking Challenge

<http://www.robocup2016.org/en/events/amazon-picking-challenge/>

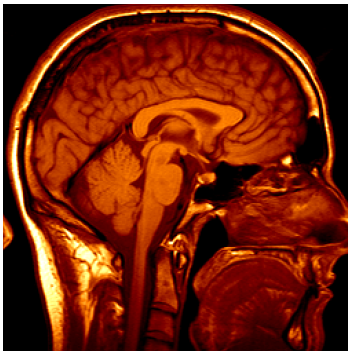


Amazon Prime Air

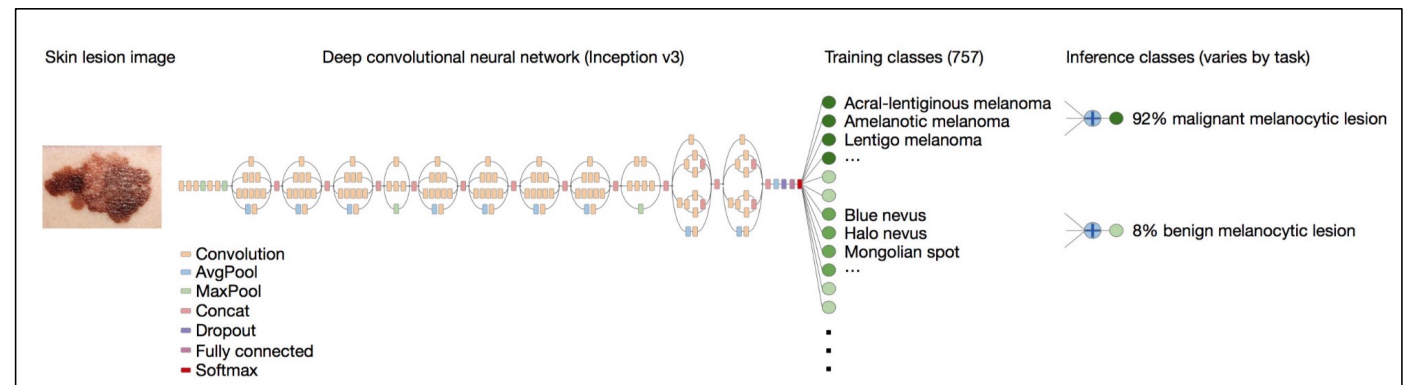


Amazon Scout

Medical imaging



3D imaging
(MRI, CT)



Skin cancer classification with deep learning
<https://cs.stanford.edu/people/esteva/nature/>

INVESTING

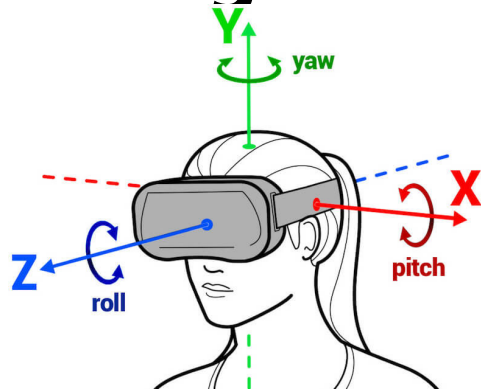
3/25/2014 @ 5:43PM | 70,399 views

Facebook Buys Oculus, Virtual Reality Gaming Startup, For \$2 Billion

[+ Comment Now](#) [+ Follow Comments](#)



Virtual & Augmented Reality



6DoF head tracking



Hand & body tracking



3D scene understanding



3D-360 video capture

Current state of the art

- You just saw many examples of current systems.
 - Many of these are less than 10 years old
- Computer vision is an active research area, and rapidly changing
 - Many new apps in the next 5 years
 - Deep learning and generative methods powering many modern applications
- Many startups across a dizzying array of areas
 - Generative AI, robotics, autonomous vehicles, medical imaging, construction, inspection, VR/AR, ...

Why is computer vision difficult?



Viewpoint variation



Illumination



Credit: Flickr user michaelpaul

Scale

Why is computer vision difficult?



Intra-class variation



Motion (Source: S. Lazebnik)

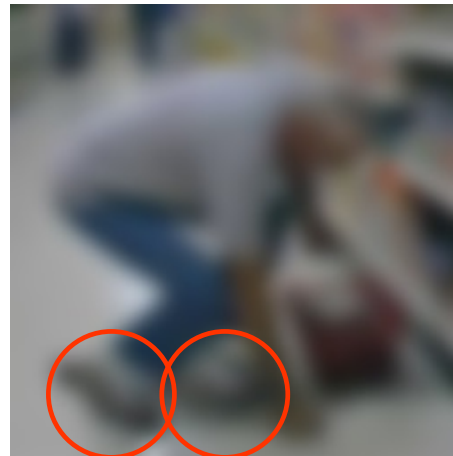
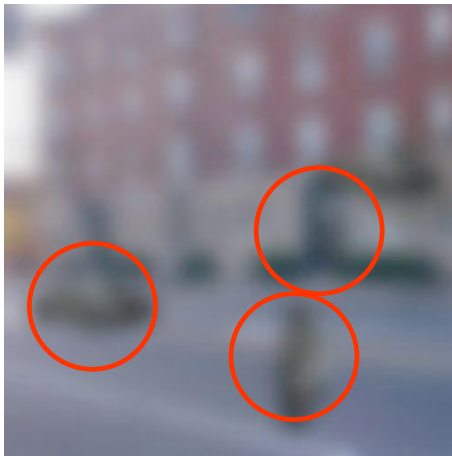
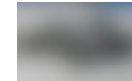
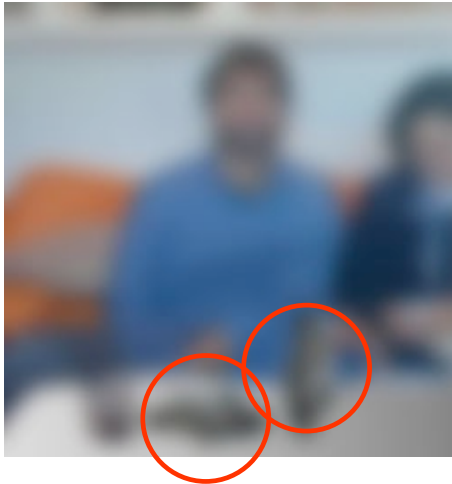


Background clutter



Occlusion

Challenges: local ambiguity



slide credit: Fei-Fei, Fergus & Torralba

But there are lots of visual cues we can use...



NATIONALGEOGRAPHIC.COM

© 2003 National Geographic Society. All rights reserved.

Source: S. Lazebnik

Bottom line

- Perception is an inherently ambiguous problem
 - Many different 3D scenes could have given rise to a given 2D image



Artist Julian Beever with his anamorphic Coke bottle

- We often must use prior knowledge about the world's structure

The state of Computer Vision and AI: we are really, really far.

Oct 22, 2012



The picture above is funny.

But for me it is also one of those examples that make me sad about the outlook for AI and for Computer Vision. What would it take for a computer to understand this image as you or I do? I challenge you to think explicitly of all the pieces of knowledge that have to fall in place for it to make sense. Here is my short attempt:

- You recognize it is an image of a bunch of people and you understand they are in a hallway
- You recognize that there are 3 mirrors in the scene so some of those people are 'fake' replicas from different viewpoints.
- You recognize Obama from the few pixels that make up his face. It helps that he is in his suit and that he is surrounded by other people with suits.
- You recognize that there's a person standing on a scale, even though the scale occupies only very few white pixels that blend with the background. But, you've used the person's pose and knowledge of how people interact with objects to figure it out.
- You recognize that Obama has his foot positioned just slightly on top of the scale. Notice the language I'm using: It is in terms of the 3D structure of the scene, not the position of the leg in the 2D coordinate system of the image.
- You know how physics works: Obama is leaning in on the scale, which applies a force on it. Scale measures force that is applied on it, that's how it works => it will over-estimate the weight of the person standing on it.
- The person measuring his weight is not aware of Obama doing this. You derive this because you know his pose, you understand that the field of view of a person is finite, and you understand that he is not very likely to sense the slight push of Obama's foot.
- You understand that people are self-conscious about their weight. You also understand that he is reading off the scale measurement, and that shortly the over-estimated weight will confuse him because it will probably be much higher than what he expects. In other words, you reason about implications of the events that are about to unfold seconds after this photo was taken, and especially about the thoughts and how they will develop inside people's heads. You also reason about what pieces of information are available to people.
- There are people in the back who find the person's imminent confusion funny. In other words you are reasoning about state of mind of people, and their view of the state of mind of another person. That's getting frighteningly meta.
- Finally, the fact that the perpetrator here is the president makes it maybe even a little more funnier. You understand what actions are more or less likely to be undertaken by different people based on their status and identity.

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—

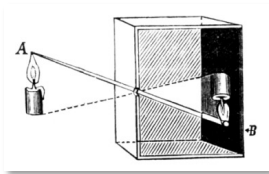
CS5670: Introduction to Computer Vision

- Project-based course whose goal is to teach you the fundamentals of computer vision – image processing, geometry, recognition – in a hands-on way
- This course covers **fundamentals, including mathematical fundamentals**. It is not a course specifically on deep learning or generative AI, though those topics will be covered.

Course requirements

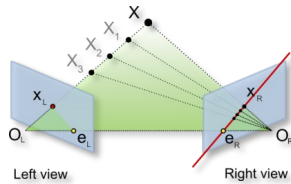
- Prerequisites
 - Data structures
 - Good working knowledge of Python programming
 - Linear algebra
 - Vector calculus
- Course does ***not*** assume prior imaging experience
 - computer vision, image processing, graphics, etc.

Course overview (tentative)



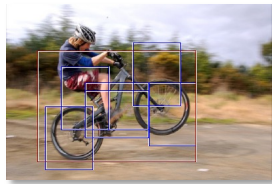
1. Low-level vision

- image processing, edge detection, feature detection, cameras, image formation



2. Geometry & appearance

- projective geometry, stereo, structure from motion, optimization, lighting & materials

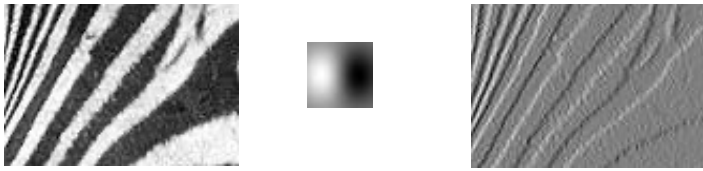


3. Recognition & generative models

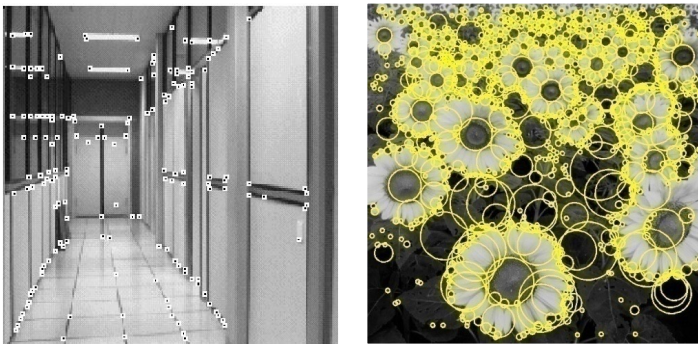
- object classification, deep learning, diffusion models

1. Low-level vision

- Basic image processing and image formation



Filtering, edge detection



Feature extraction

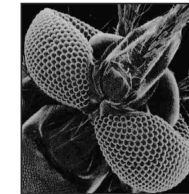
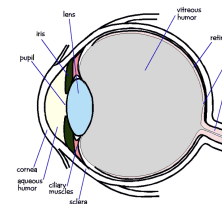
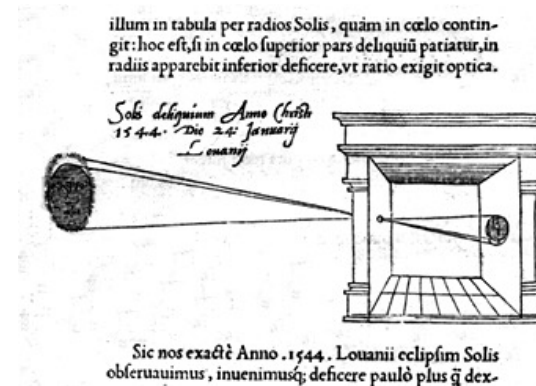
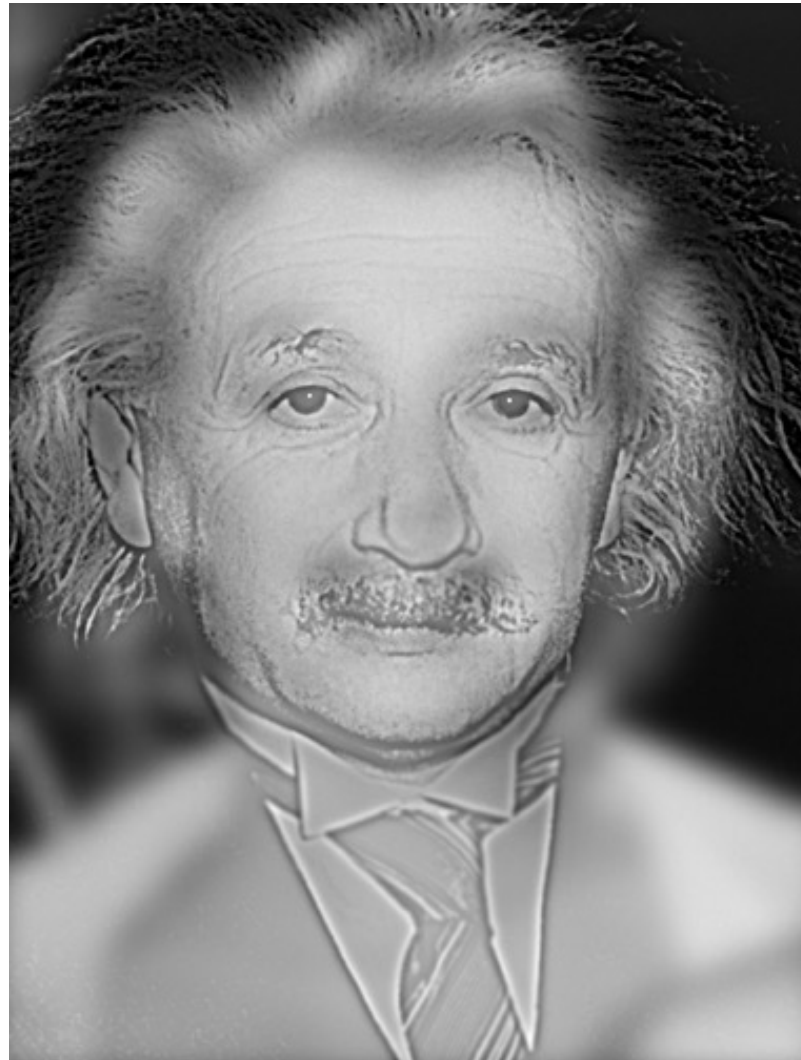


Image formation

Project: Hybrid images



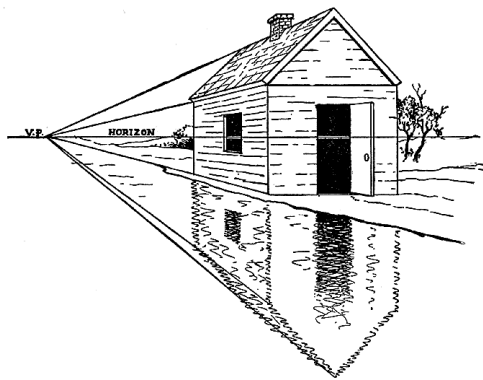
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Project: Feature detection and matching



2. Geometry & appearance



Projective geometry

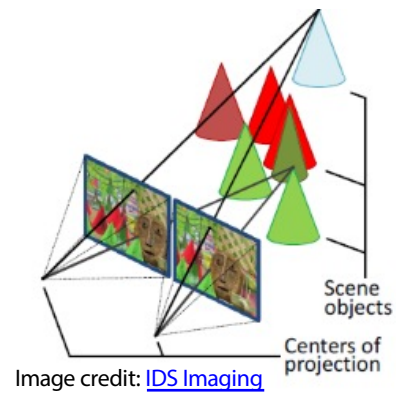
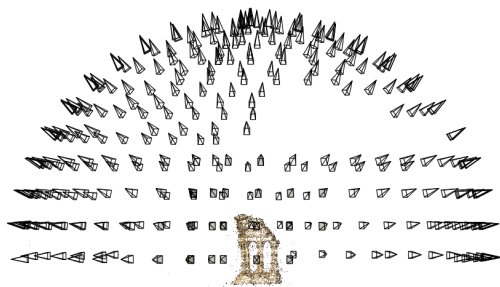


Image credit: [IDS Imaging](#)

Stereo vision



Multi-view stereo



Structure from motion

Project: Creating panoramas



Project: Neural Radiance Fields (NeRFs)

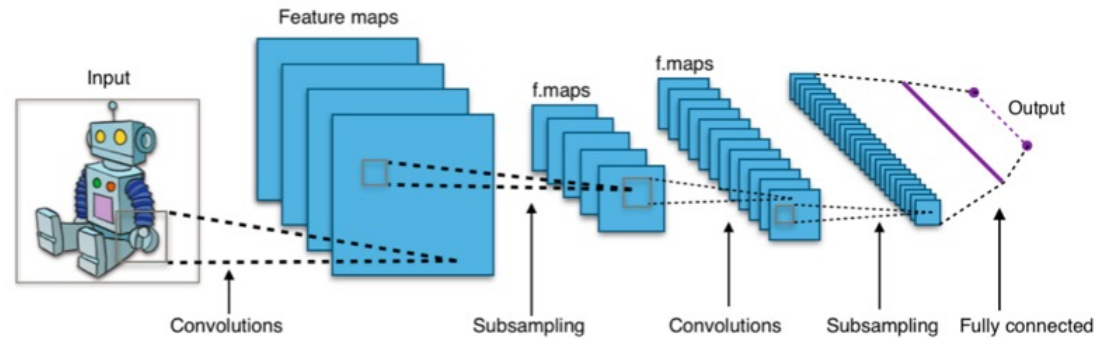


3. Recognition, Deep Learning & Generative Models



→ “dog”

Image classification



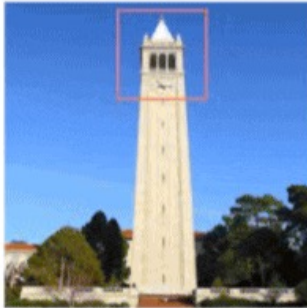
Convolutional Neural Networks



“a class watching a computer vision lecture at Cornell Tech”

Image generation

Project: Image diffusion models



Hole Filling



"Make it Real"



A Lithograph of a Waterfall



An Oil Painting of an Old Man

Lectures

- Lectures will be held in person in Bloomberg 131
- If there is an instance where you need to attend lecture remotely, please reach out to the instructor for approval
- No conversations during class, please!

Grading

- Approximately weekly short quizzes (typically at the beginning of class on Thursdays)
- One midterm (take-home), one final exam (in class)
- Grade breakdown (subject to minor tweaks):
 - Quizzes: 5% (lowest quiz grade dropped)
 - Midterm: 16%
 - Programming projects: 63%
 - Final exam: 16%

Late policy

- Four free “slip days” will be available for the semester
- A late project will be penalized by 10% for each day it is late (excepting slip days), and no extra credit will be awarded

Academic Integrity

- Assignments will be done solo or in pairs (we'll let you know for each project)
- Please do not leave any code public on GitHub (or the like) at the end of the semester!
- We will follow the Cornell Code of Academic Integrity (<http://cuinfo.cornell.edu/aic.cfm>)
- If you use ChatGPT (or CoPilot, or similar) on coding assignments, you must disclose that with your submission
 - **BUT:** We advise you to do all coding yourself, unassisted. You will learn less, and become less capable experts in vision, if you rely on LLMs.

Questions?